

[01101>

QUEPFL

QUANTUM ICING

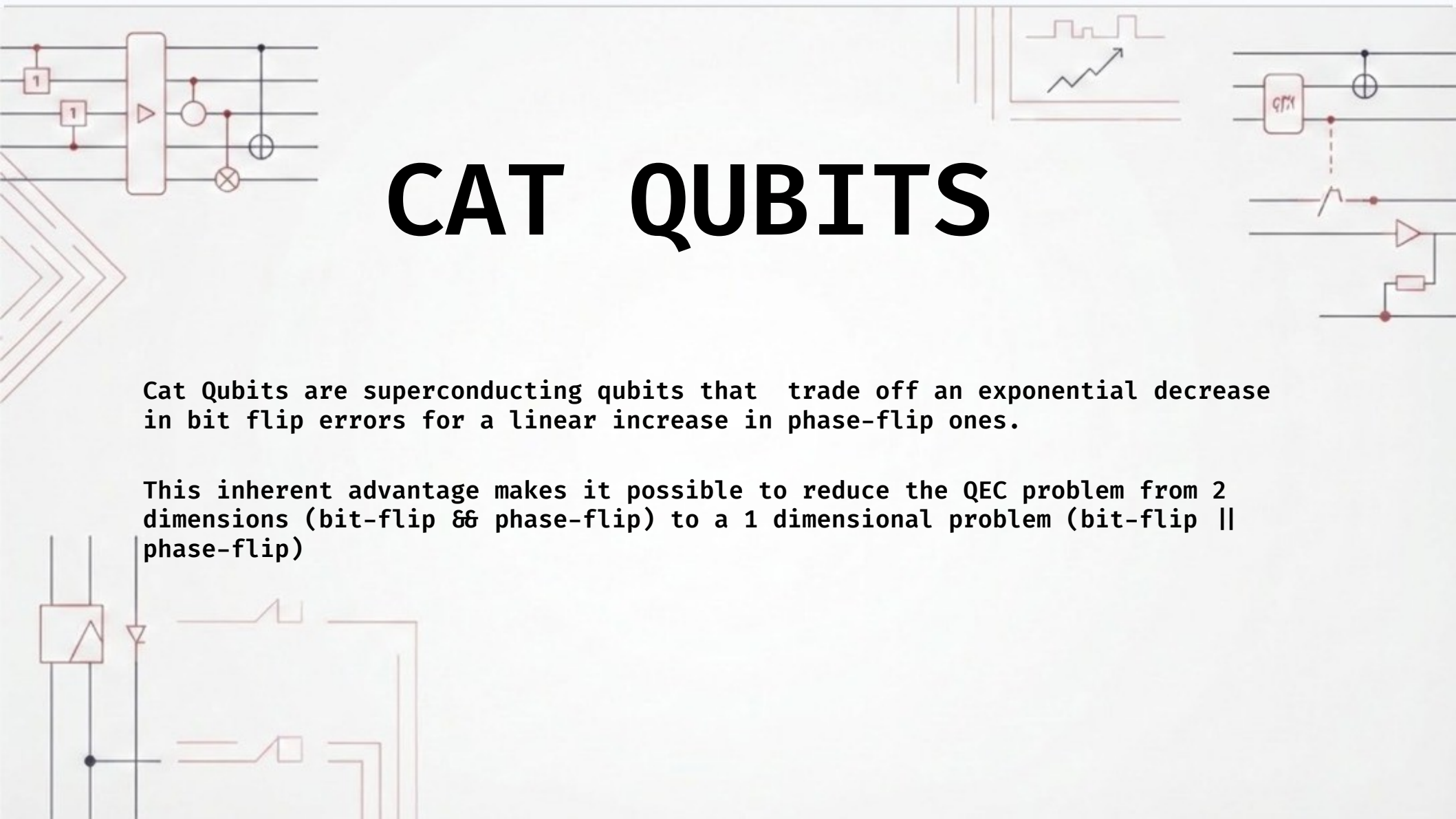
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QPFL HACKATON 2026



BIT

Lausanne, Switzerland 2026



CAT QUBITS

Cat Qubits are superconducting qubits that trade off an exponential decrease in bit flip errors for a linear increase in phase-flip ones.

This inherent advantage makes it possible to reduce the QEC problem from 2 dimensions (bit-flip & phase-flip) to a 1 dimensional problem (bit-flip || phase-flip)

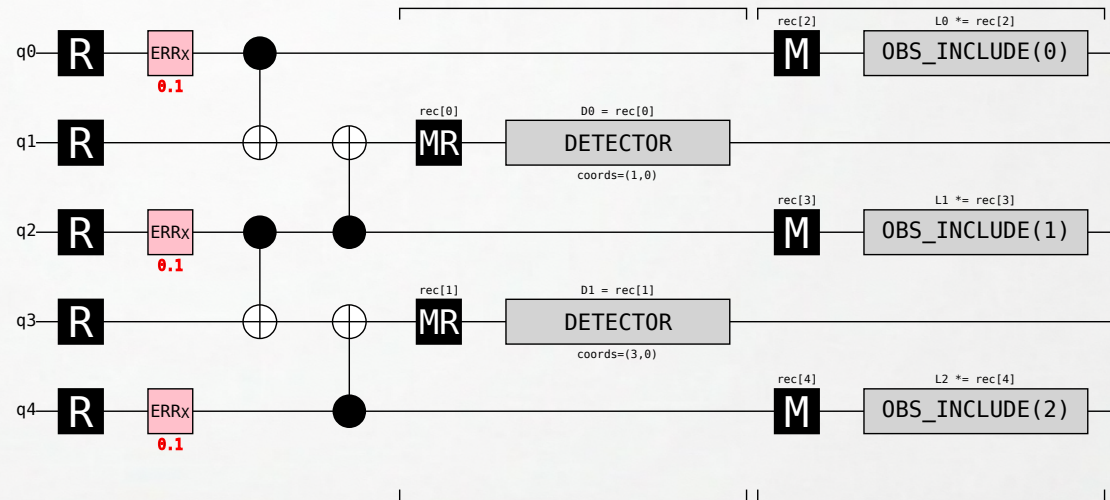
HOW DID WE USE AI?



Task 3: Repetition Code and Decoder

The three ingredients:

- an encoding tuple (n, k, d)
- stabilizers
- decoder



Repetition Code Experiment: Memory Mode Threshold

- GOAL



Estimate logical error probability of **repetition code** in memory mode & **identify threshold**

- NOISE MODEL



Phenomenological **bit-flip** modeled as an X gate with probability p

- DECODING

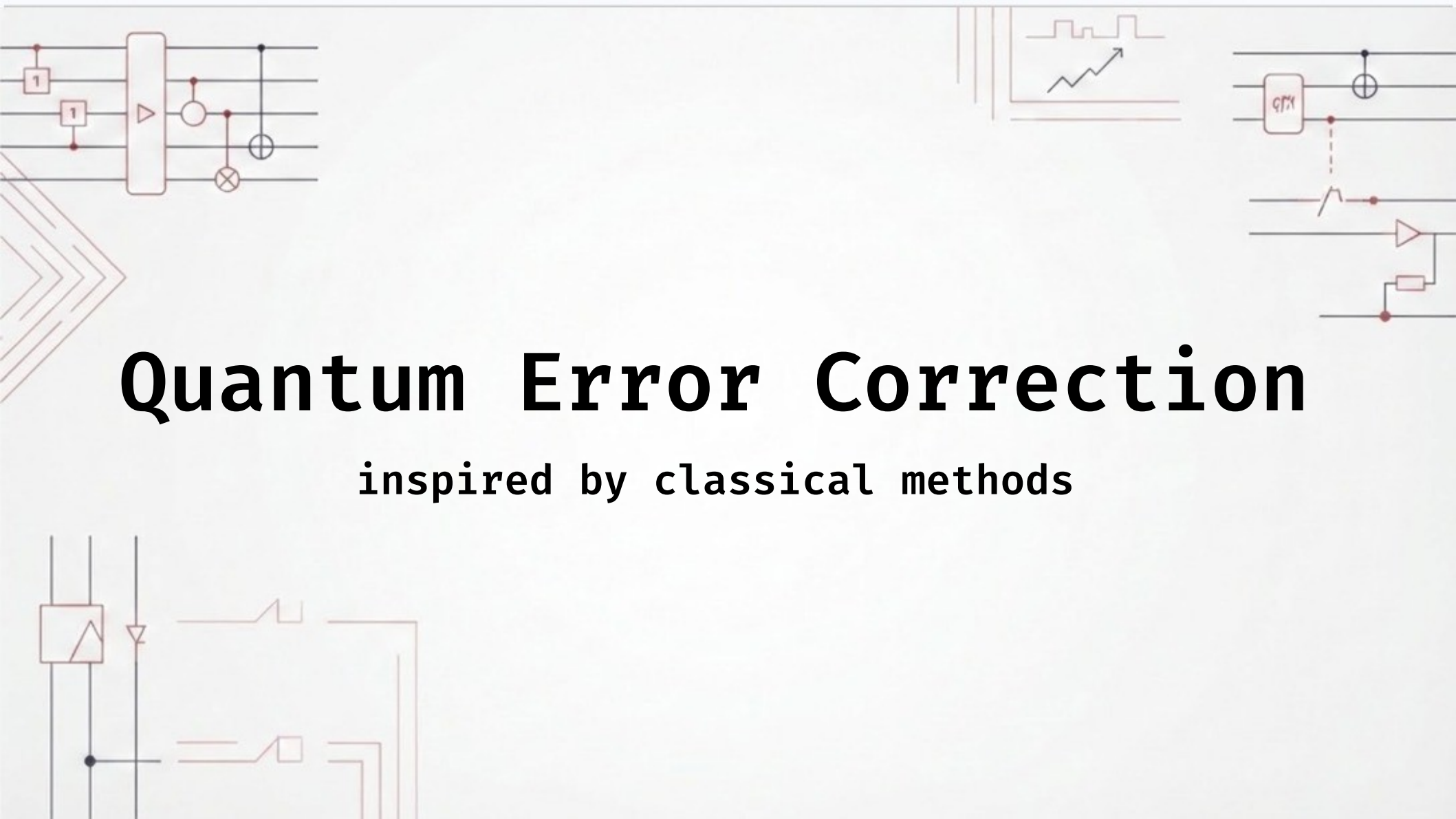


Decoding before detectors

- NUMERICAL ESTIMATE



Parametrization over error probability p and distances d

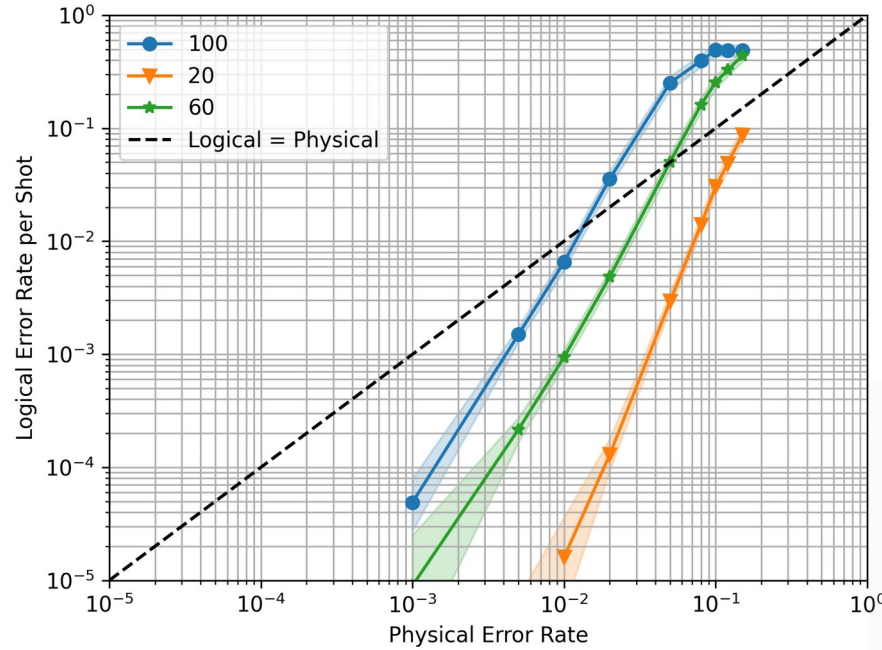
The background features several faint, stylized quantum circuit diagrams. In the top left, a circuit with three horizontal lines includes two red square gates labeled '1' and a red triangle gate. In the top right, a circuit shows a red square gate labeled 'CNOT', a red circle with a plus sign, and a red square gate. In the bottom left, a circuit includes a red square gate with a triangle inside, a red triangle gate, and a red square gate. In the bottom right, a circuit shows a red square gate, a red triangle gate, and a red square gate. A red line graph with an upward arrow is also visible in the top right.

Quantum Error Correction

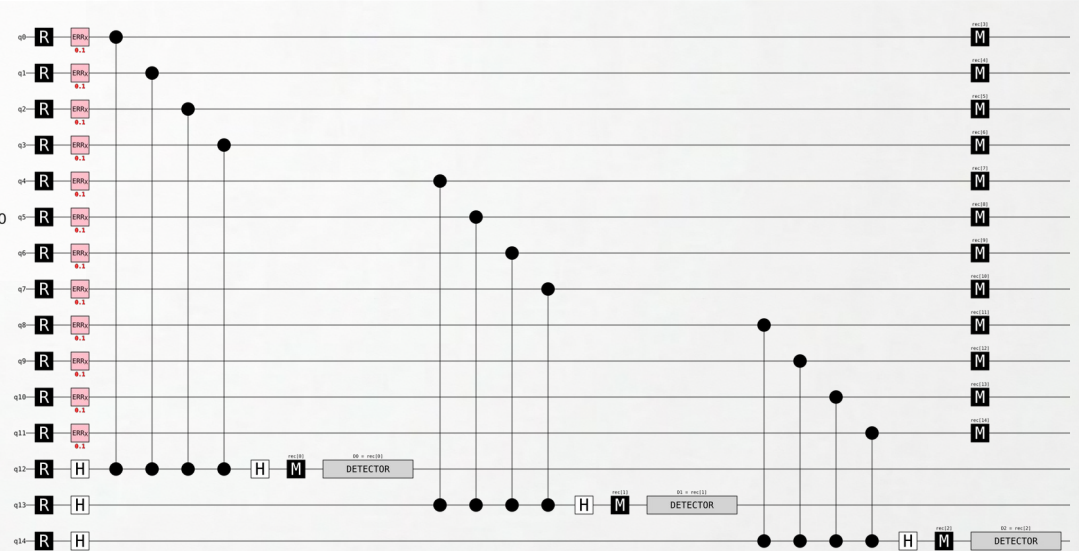
inspired by classical methods

Low-Density Parity Codes

A brave experiment



AN INTUITIVE IMPLEMENTATION OF LDPC



Custom H

(made by us, 2 forms)

Format A (1 ancilla per 4 data + 2 global)

[1 1 1 1 0 0 0 0] # ancilla 1

[0 0 0 0 1 1 1 1] # ancilla 2

[1 1 0 0 1 1 0 0] # global 1

[1 0 1 0 1 0 1 0] # global 2

Easily scales by adding 1 row, 4 column.

Added row (before globals) contains 0s – 1111

Global rows repeat the “1100” / “1010” pattern in the new columns.

Rate grows towards $4/5 \sim 0.8$

Threshold $\sim 10^{-2}$

Thresholds lowers for larger H

Format B (3 ancillas per 4 data)

[1 1 1 1 0 0 0 0]

[1 1 0 0 0 0 0 0]

[1 0 1 0 0 0 0 0]

[0 0 0 0 1 1 1 1]

[0 0 0 0 1 1 0 0]

[0 0 0 0 1 0 1 0]

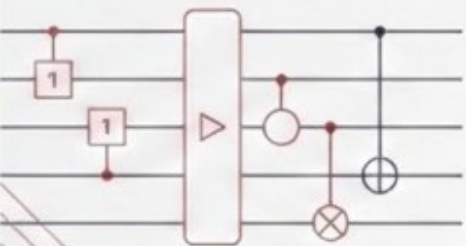
Easily scales by adding 3 rows, 4 columns

Added rows contain the pattern along diagonal.

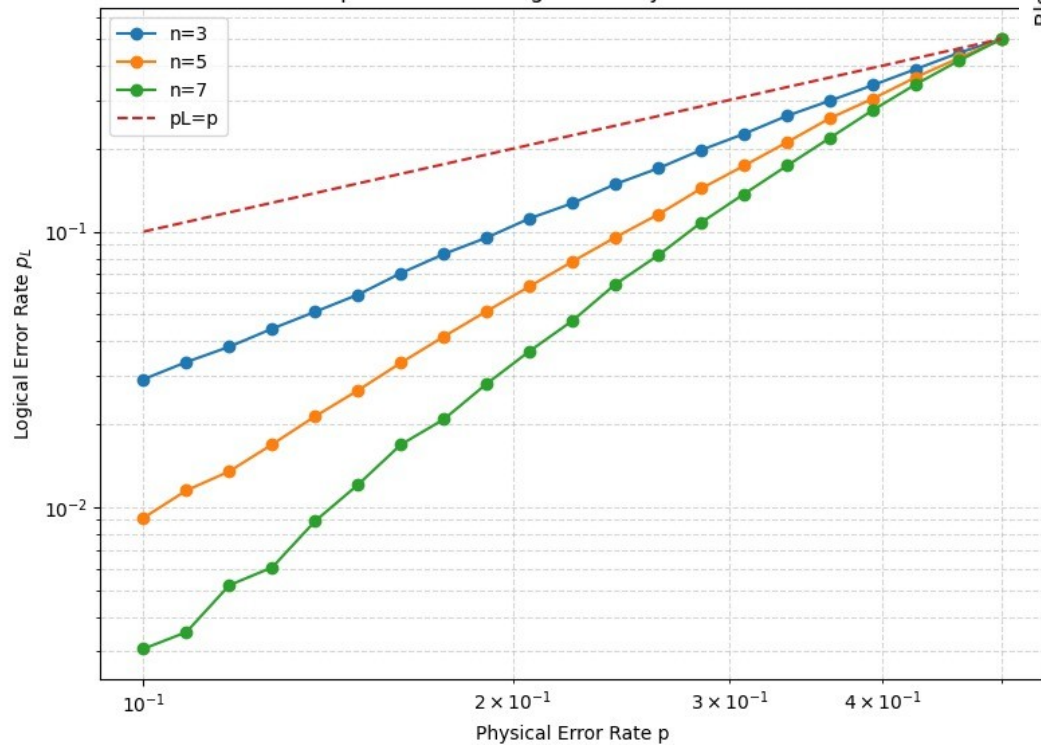
Rate fixed at $4/7 \sim 0.57$

Threshold $\sim 10^{-1}$

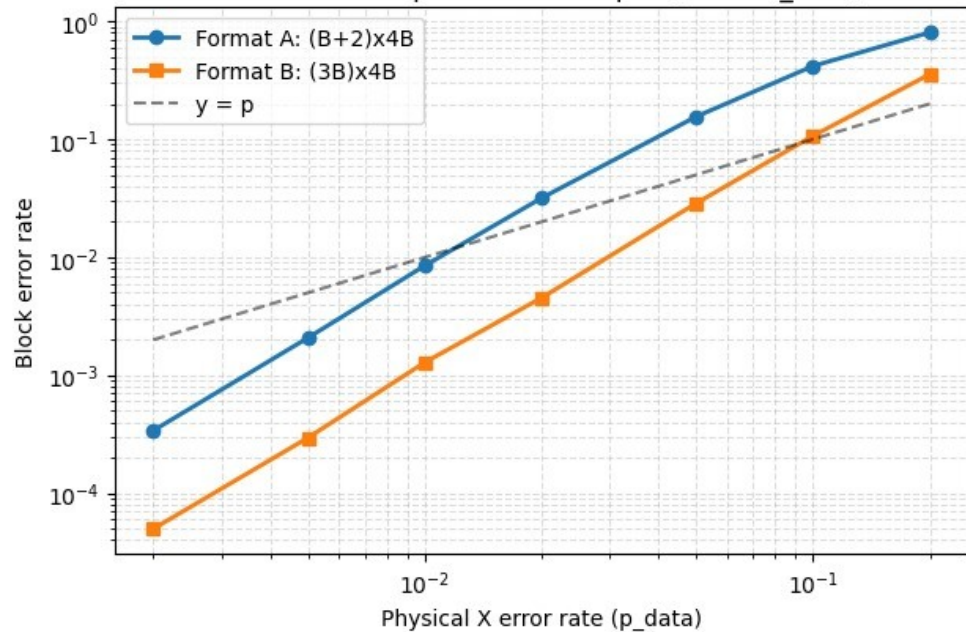
Thresholds lowers for larger H



Repetition Code: Logical vs Physical Error Rate



Custom H comparison on one plot (same n_{data})



Binary Golay

The *perfect* linear code

Punctured Binary Golay

$[n=23, k=12, d=7]$ -code

$$t = \lfloor \frac{7-1}{2} \rfloor$$

(correct up to 3 bit-flips).

Perfect code: every syndrome maps to a unique error with $wt \leq 3$ (full syndrome table). Block failure for independent bit-flips p :

$$P_{fail}(p) = 1 - \sum_{i=0}^3 \binom{23}{i} p^i (1-p)^{23-i}$$

Reference: Raymond Hill, *A First Course in Coding Theory*, Oxford University Press, 1986.

Extended Binary Golay

[n=24, k=12, d=8]-code

$$t = \lfloor \frac{8-1}{2} \rfloor$$

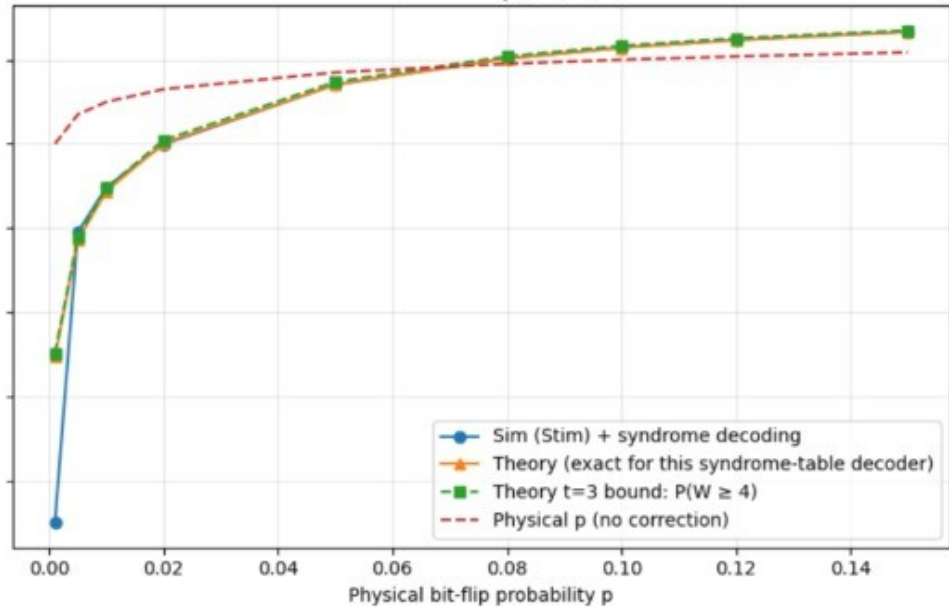
(same bounded-distance correction radius).

Advantage: larger distance (better separation / robustness), cost: +1 qubit per block. Block failure (independent bit-flips p):

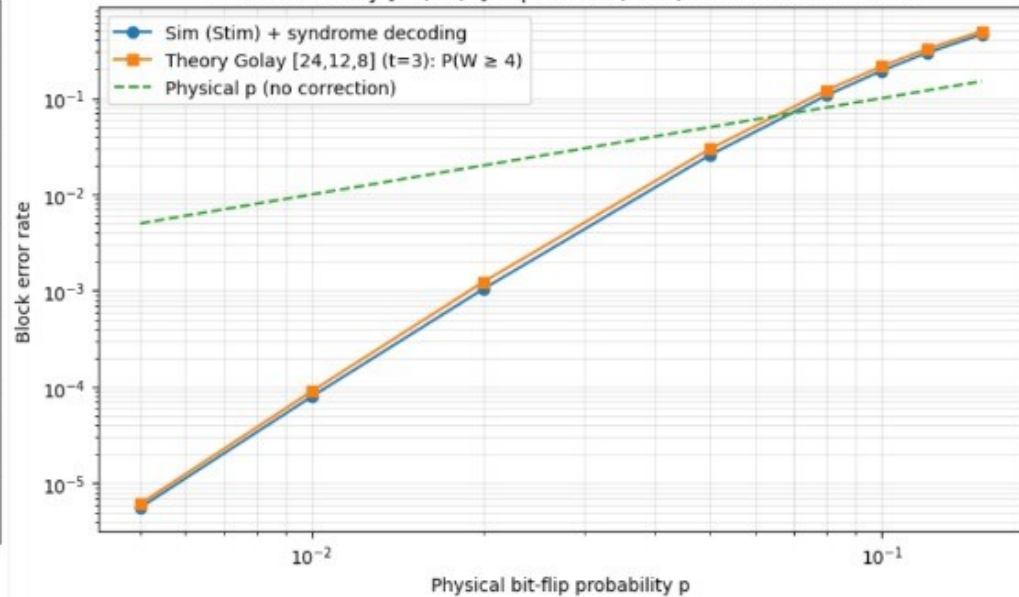
$$P_{fail}(p) = 1 - \sum_{i=0}^3 \binom{24}{i} p^i (1-p)^{24-i}$$

Results

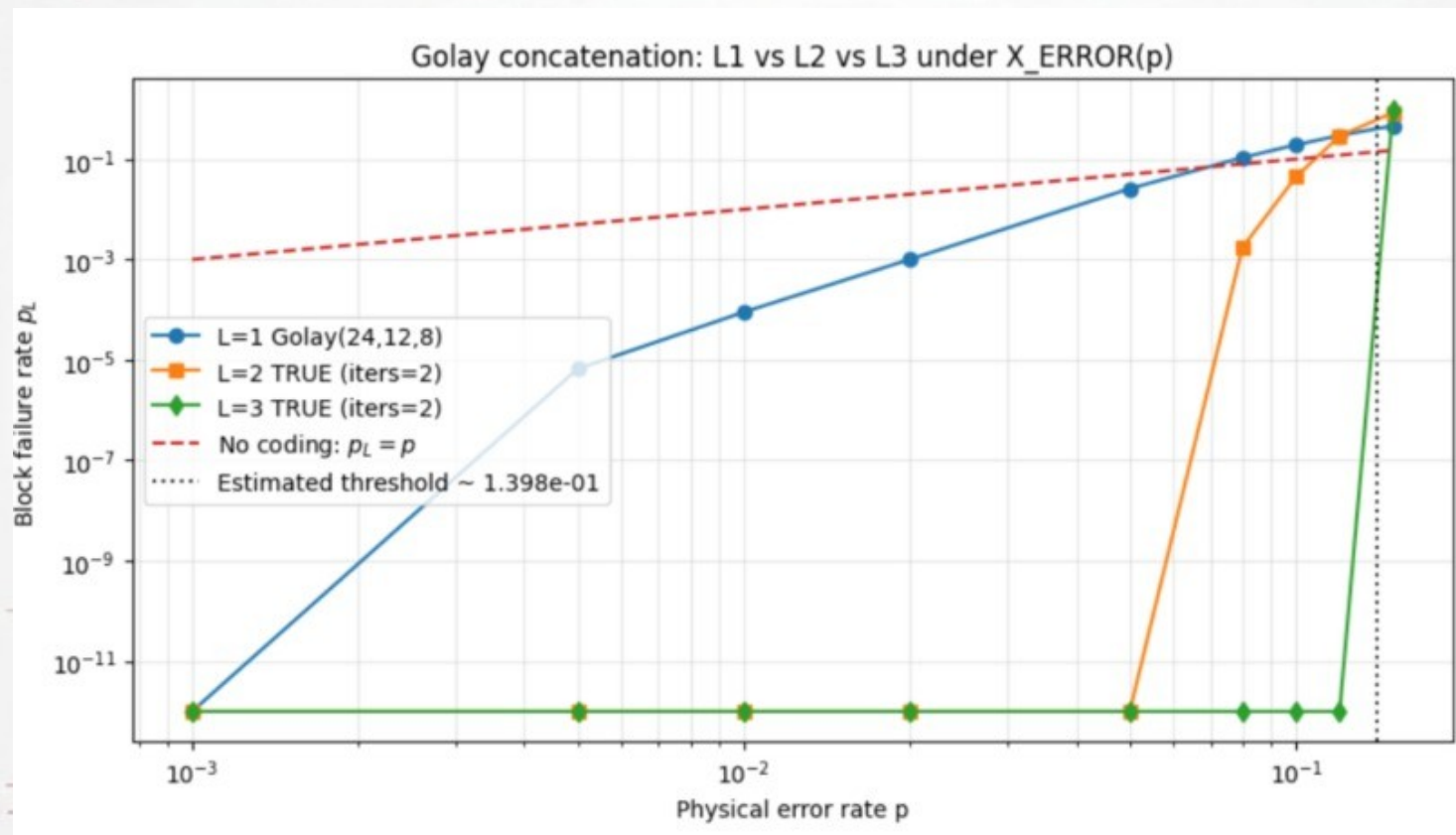
Extended Golay [24,12,8]



Extended Golay [24,12,8] - quantum (Stim) vs theoretical bound

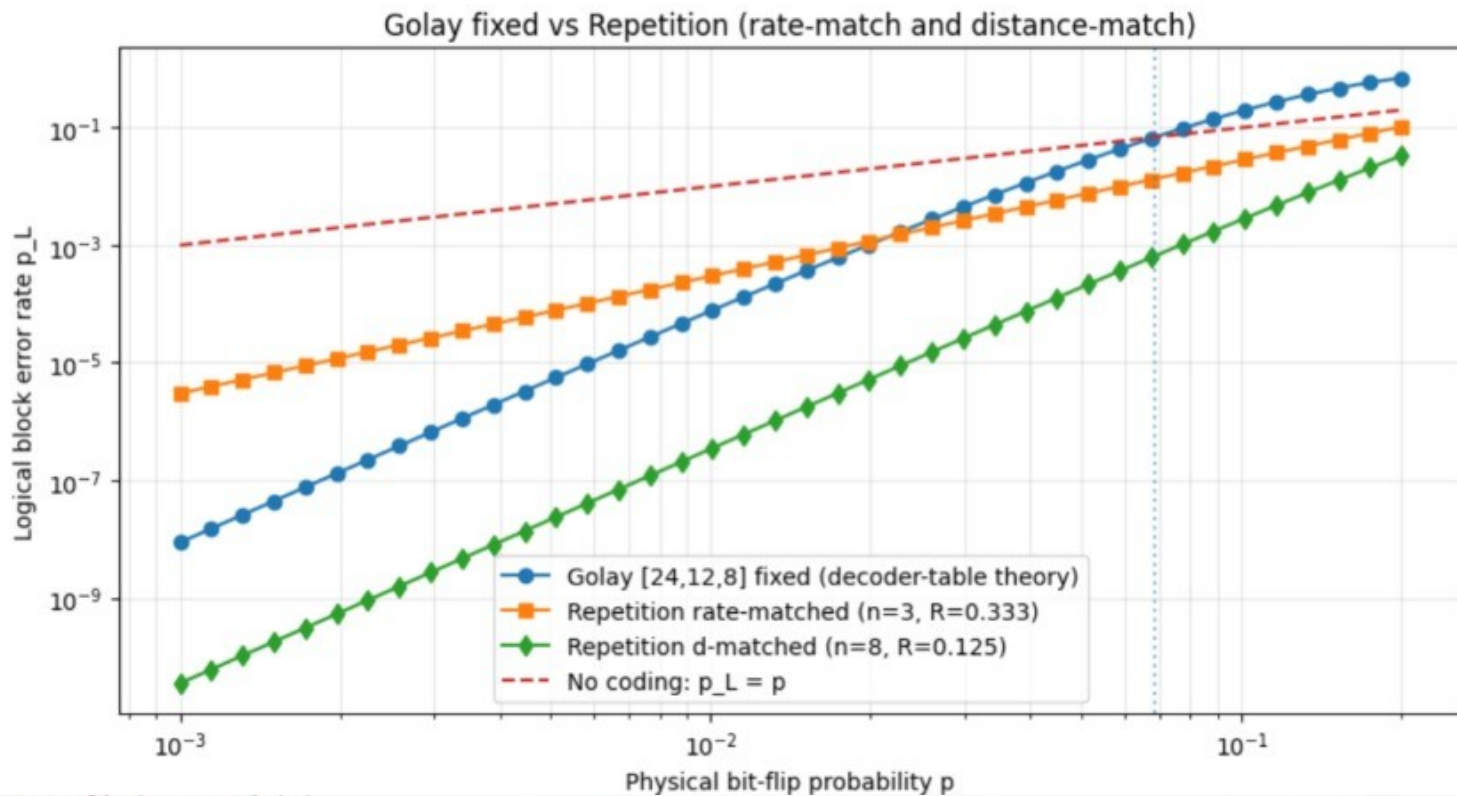


Golay 3 Level concatenation

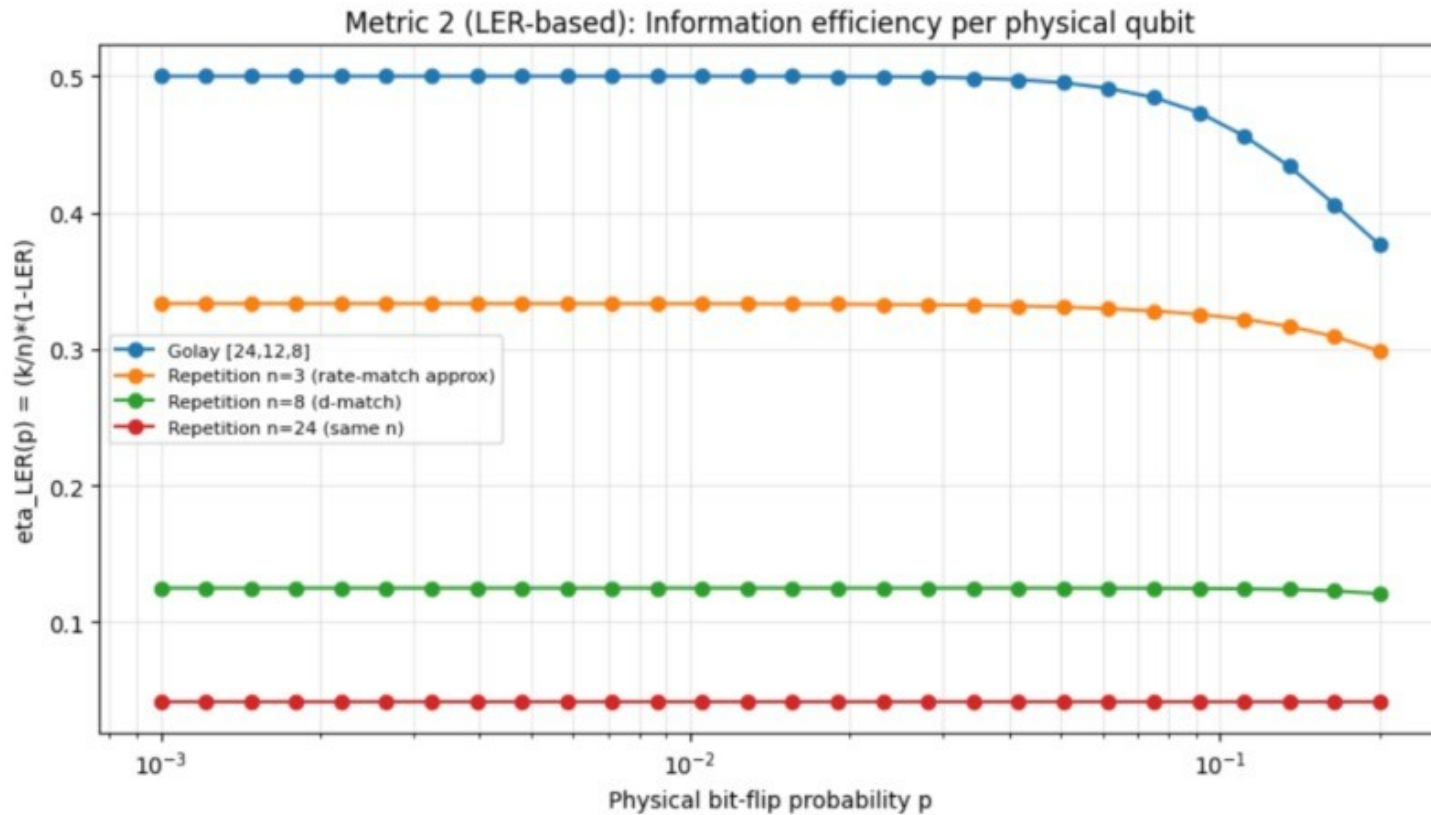


[illegible]

Golay vs Repetition



Golay vs Repetition



Steane Overview

Parameters: $n=7$, $k=1$, $d=3$

CSS Architecture: Separates X and Z tracking to analyze hardware noise bias.

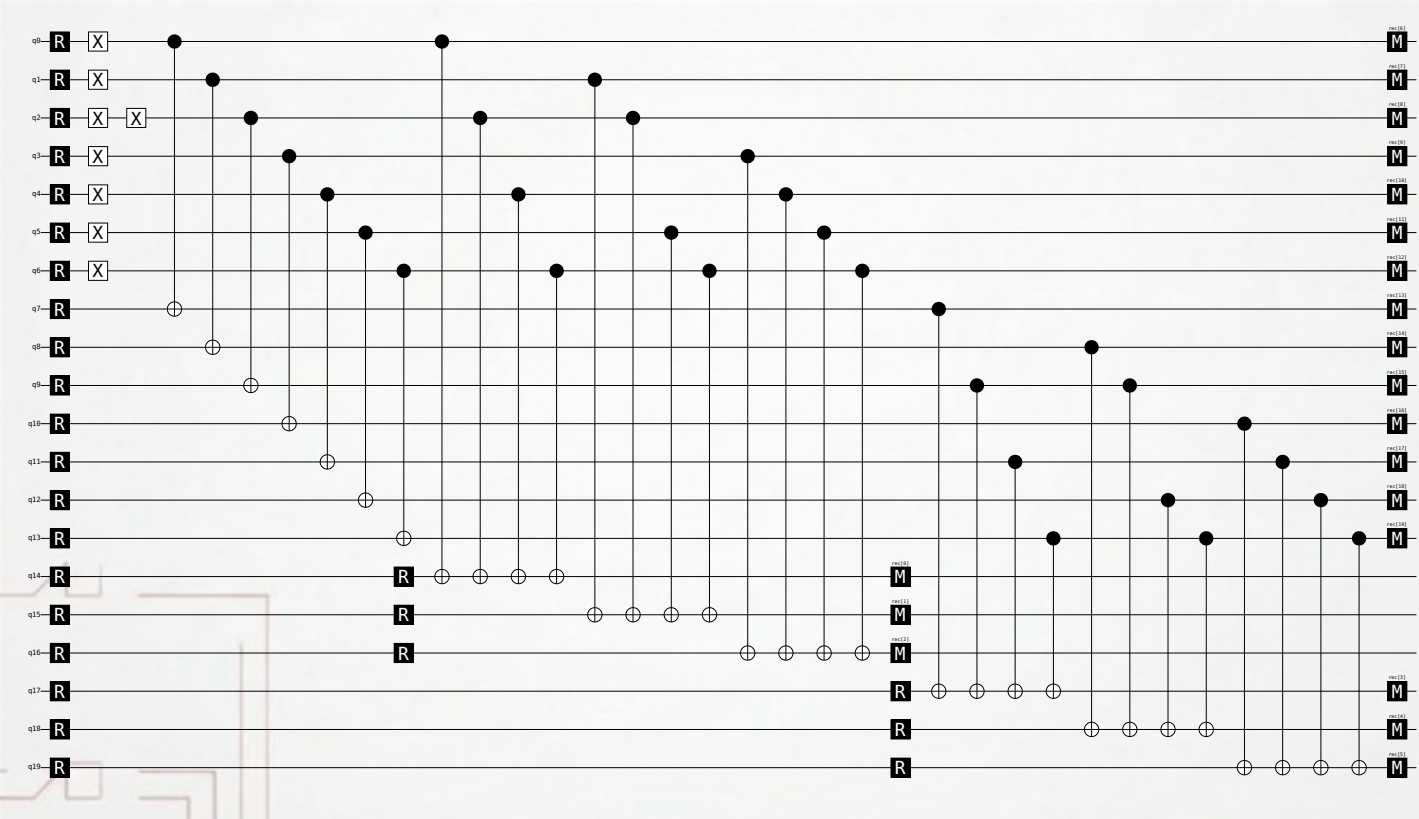
Deterministic Prep: Encodes perfect 0 states without dropping shots.

Safe Extraction: Non-destructive parity checks using dedicated ancilla qubits.

Precise Decoding: Deterministic Hamming-mapped decoder to track physical errors instantly.

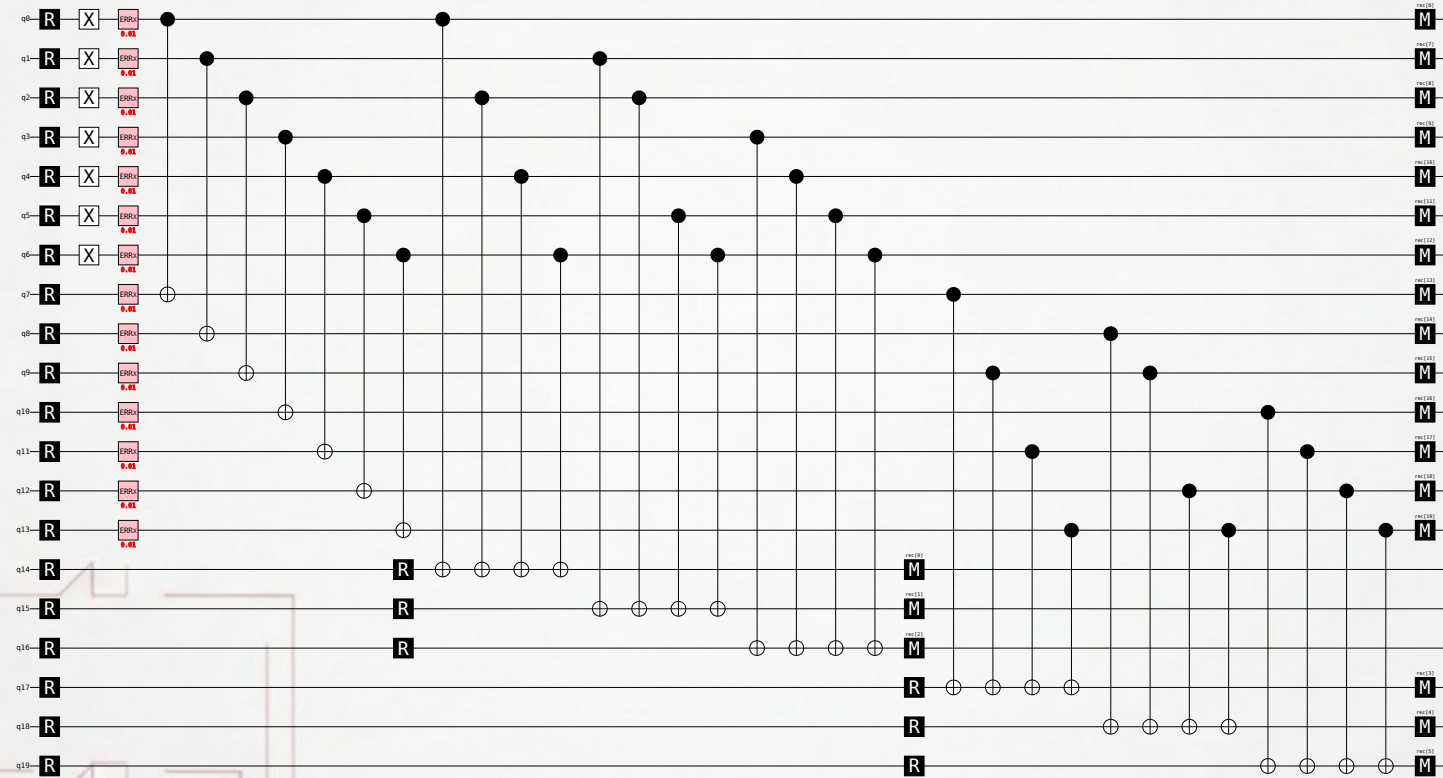
Steane

CNOT - scenario A



Steane

CNOT - scenario B



Steane CNOT overview

The Math: Propagates X to X from control to target.

The Advantage: Never rotates X errors into the lethal Z basis.

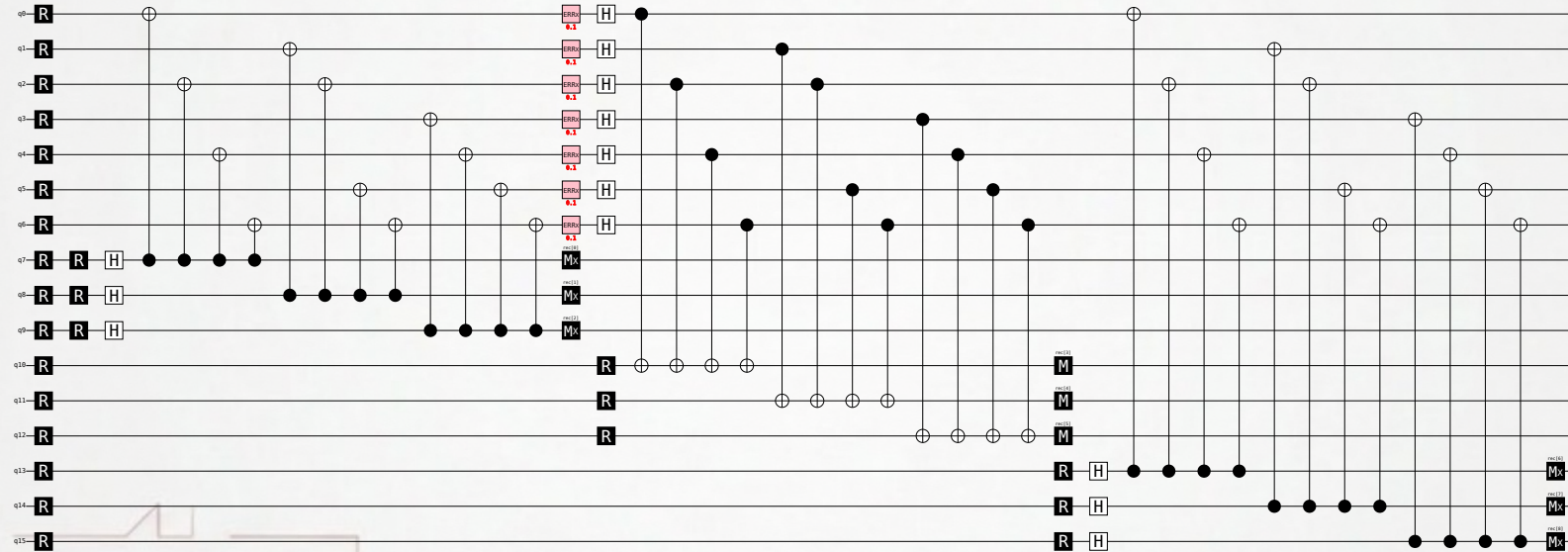
Simulation Error Rate:

- Scenario A: 0% error
- Scenario B: 0.000075% ($p = 0.01$, shots = 20000)

Verdict: CNOT is natively safe. We must use it via **Gate Teleportation** to safely execute H and S gates.

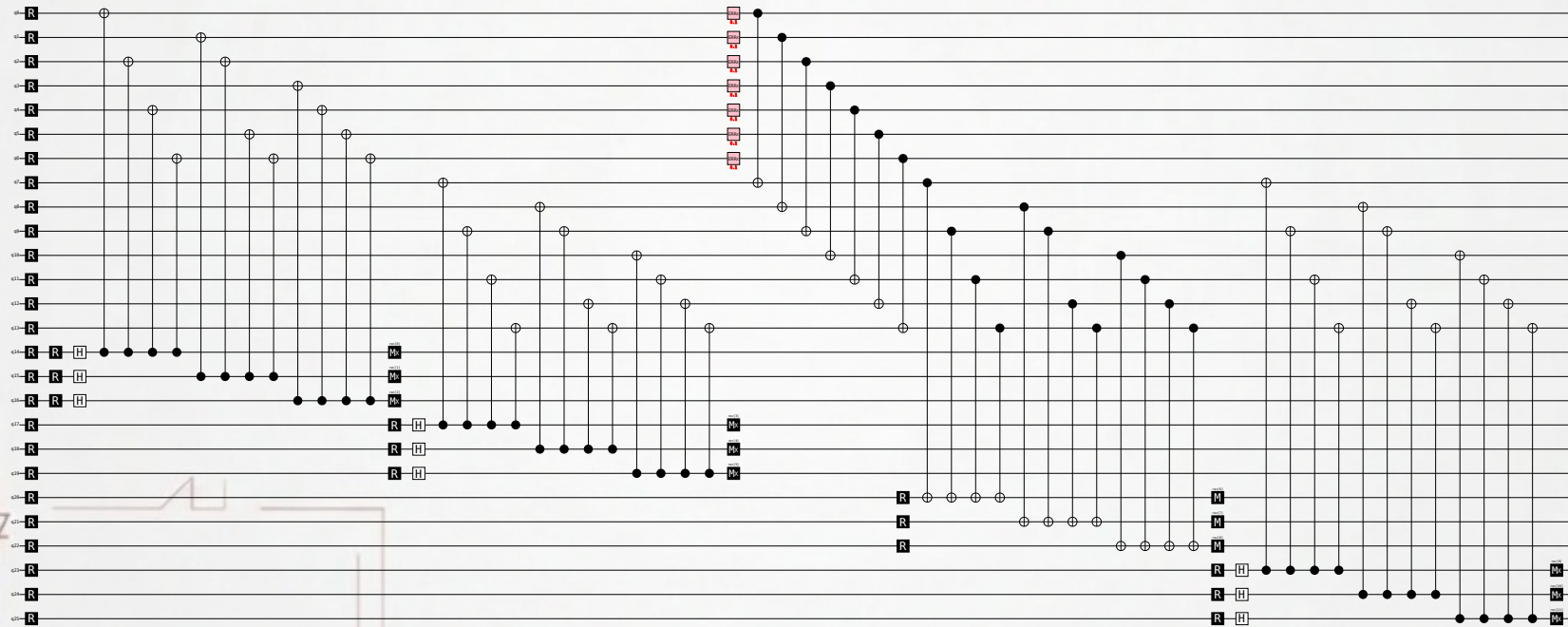
Steane

H - scenario A



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H - scenario B



Steane H overview

Scenario A: Direct Transversal H Gate

~51.87% → Z-Error Triggers

Comment: The H gate mathematically swaps axes ($X \rightarrow Z$).

Harmless bit-flips are converted into lethal phase-flips, completely destroying Cat Qubit bias.

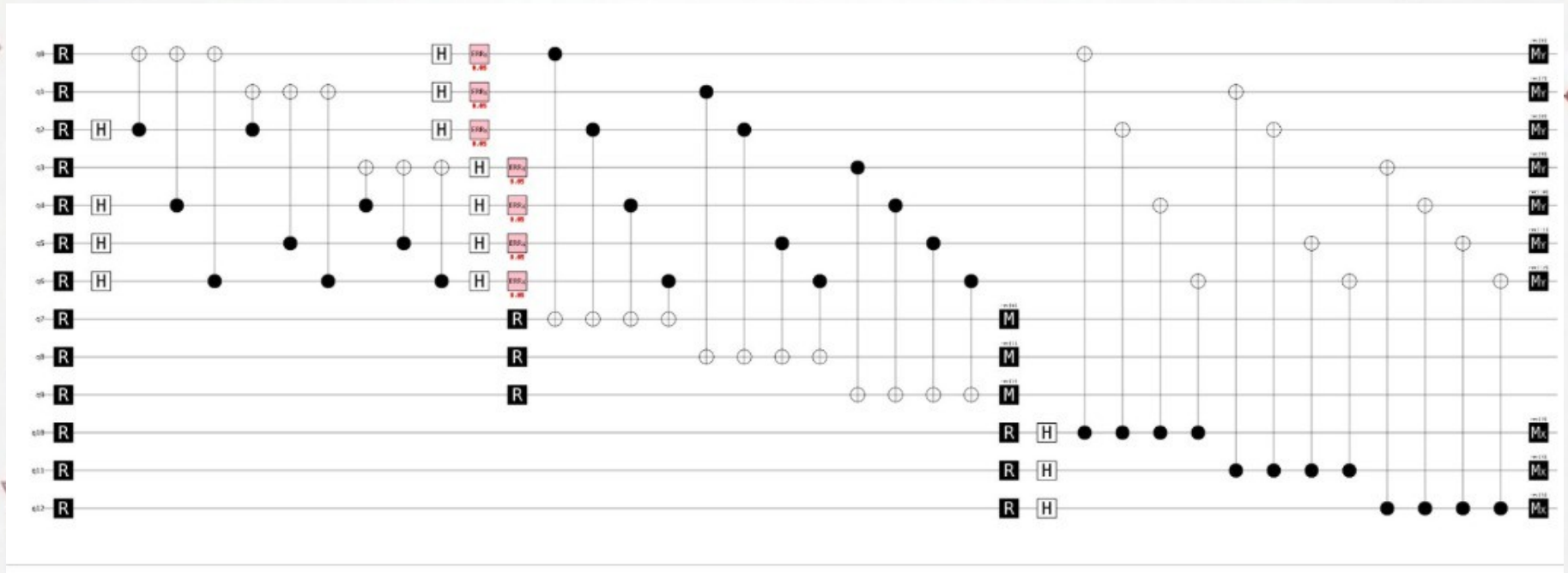
Scenario B: CNOT-Based Gate Teleportation

~50.15% → X-Error Triggers

Comment: Errors safely propagate strictly as bit-flips ($X \rightarrow X$).

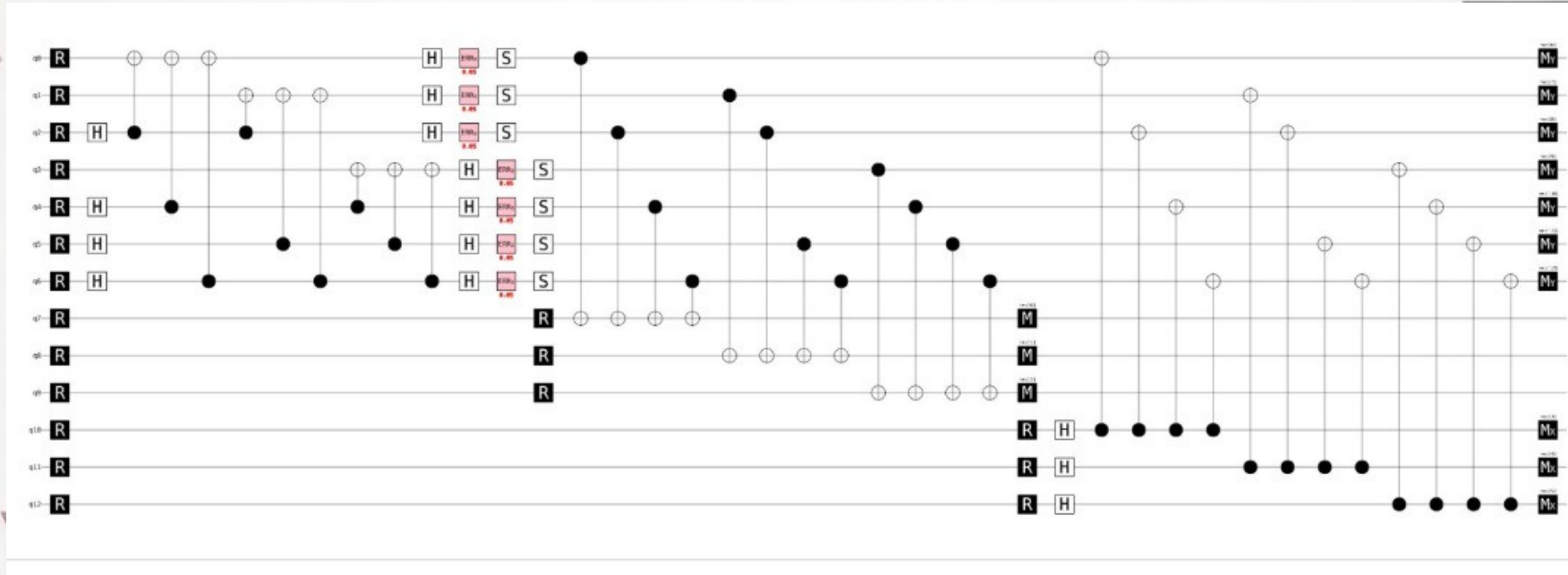
Steane

S - scenario A



Steane

S - Scenario B



Steane S overview

Scenario A: Direct Transversal H Gate

~30.00% → X-Error Triggers

0.00% → Z-Error Triggers

Scenario B: CNOT-Based Gate Teleportation

~30.00% → X-Error Triggers

~30.00% → Z-Error Triggers

Errors safely propagate strictly as bit-flips ($X \rightarrow X$).

Golay Gates

Logical meaning: with the common “extended Golay as 12×24 CSS checks” setup you often end up with $k = 0$, so you’re showing stabilizer/syndrome behavior, not a gate on a logical qubit (unless you move to quantum Golay $[[24,1,8]]$).

Syndrome extraction cost: Golay checks are heavier/denser, so measuring stabilizers needs many more CNOTs + ancillas, making error propagation and scheduling much more important than in small codes.

Baseline handling: Golay demos often require reference rounds / syndrome differencing (especially for X-checks), otherwise “syndrome triggers” can be dominated by a nontrivial baseline rather than noise or the gate effect.

Decoding burden: Golay decoding is not a tiny lookup; you typically need a more structured decoder to interpret syndromes reliably.

THANK YOU

Q&A

